

CHEM 2312 LS

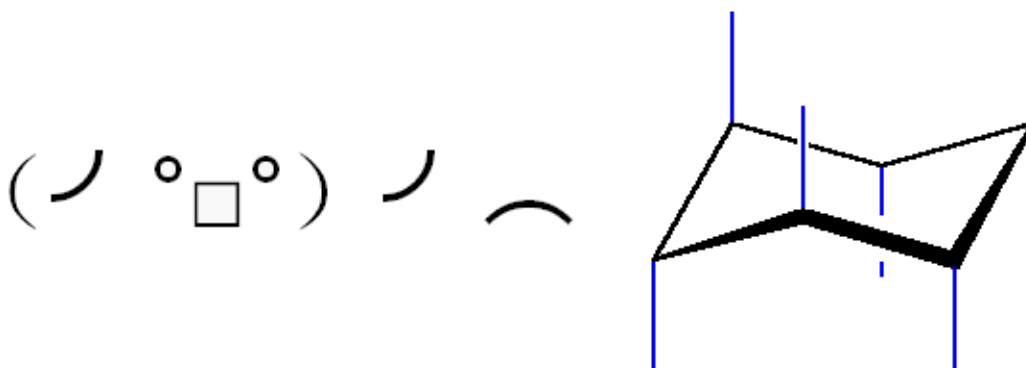
Organic Chemistry II

Midterm Examination 3

Take a deep breath. You got this.

Do not open this exam until you are instructed to do so. Only a pencil or pen, scratch paper, and molecular model kit are permitted. You will have 50 minutes to complete the exam; it can be submitted at any time after 25 minutes have elapsed. Write all responses clearly in the boxes or spaces provided; work outside of designated areas will not be graded. On multiple-choice questions, write the letter of your choice in the box *and* circle your choice.

Ensure that configurations can be unambiguously assigned at all stereocenters (*R/S* or *E/Z*). Use “+ En” to indicate the presence of a racemic mixture when chiral products are generated from achiral starting materials.

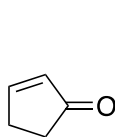


Name

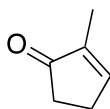
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Multiple-choice Questions. Select the best response. Circle your choice and write its letter in the box. Each question is worth **4 points**.

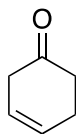
1. Which of the following enones *cannot* be prepared directly from a dicarbonyl compound via intramolecular aldol condensation?



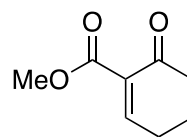
I



II



III

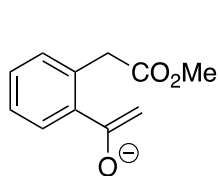
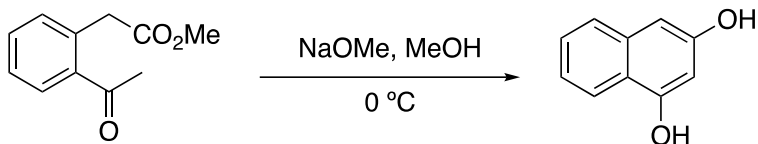


IV

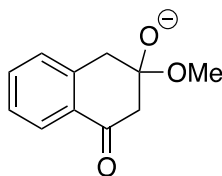
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- A. Compound I
B. Compound II
C. Compound III
D. Compound IV
E. All compounds can be prepared using intramolecular aldol condensation.

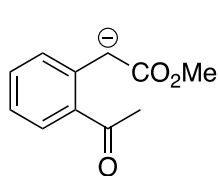
2. Consider the mechanism of the Claisen-type condensation below. Which of the following answer choices correctly orders the reactive intermediates involved in this mechanism? Note that one of the structures below is not involved!



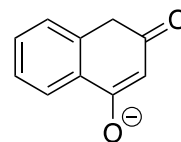
I



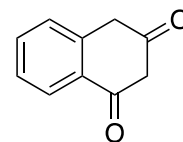
II



III



IV

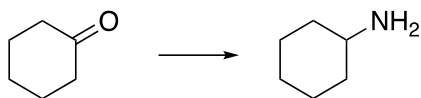


V

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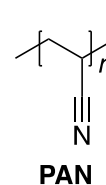
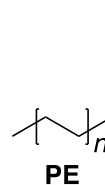
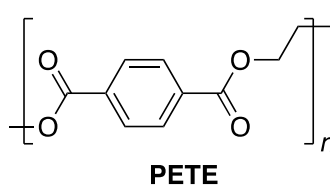
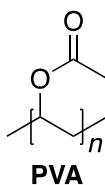
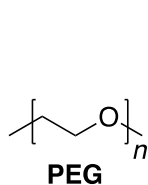
- A. I $\rightarrow$ II $\rightarrow$ V $\rightarrow$ IV
B. I $\rightarrow$ III $\rightarrow$ II $\rightarrow$ V
C. III $\rightarrow$ II $\rightarrow$ IV $\rightarrow$ V
D. III $\rightarrow$ I $\rightarrow$ V $\rightarrow$ IV
E. I $\rightarrow$ II $\rightarrow$ III $\rightarrow$ V

3. Which of the following sets of reaction conditions will accomplish the transformation below in highest yield?

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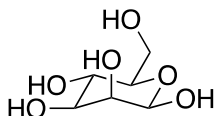
- A. NH_3 , HOAc
B. 1. NH_4Cl ; 2. LiAlH_4
C. 1. NaNH_2 ; 2. H_2 , Pd
D. 1. NH_4Cl ; 2. $\text{Na}(\text{BH}_3\text{CN})$
E. 1. LiAlH_4 ; 2. H_2O ; 3. PBr_3 ; 4. NH_3

4. Which of the following polymers is best made through a polycondensation reaction?

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- A. PEG
B. PVA
C. PETE
D. PE
E. PAN

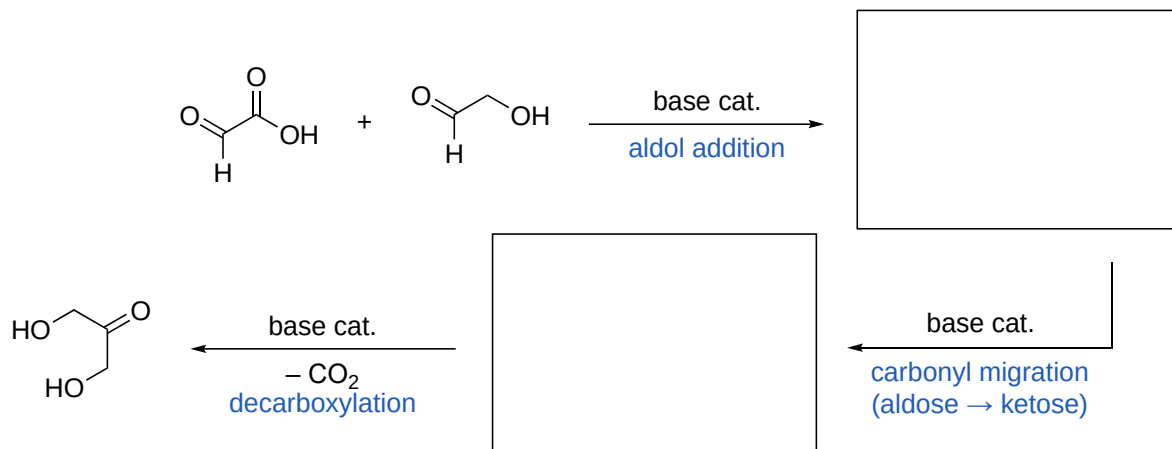
5. Which of the following statements concerning the monosaccharide below is *true*?

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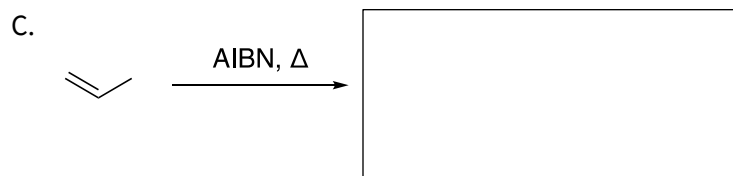
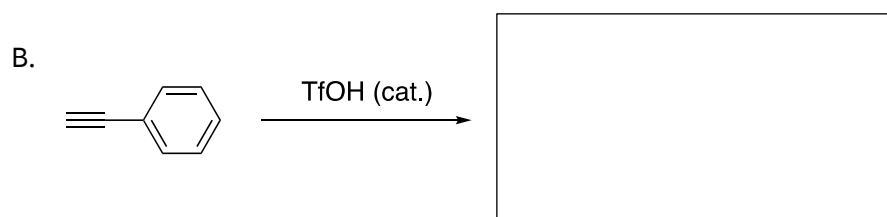
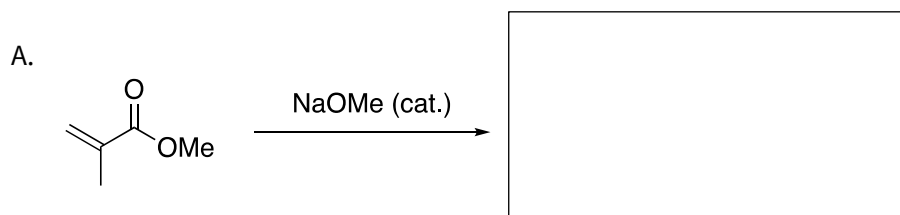
- A. The structure shown is a furanose.
B. The monosaccharide is an L-sugar.
C. The “flipped” chair conformer is more stable than the one shown.
D. The structure shown is an aldopentose.
E. The structure shown is more stable than its anomer.

Other Questions. Place your response in the designated space for each question.

6. **8 pts.** Georgia Tech's own Dr. Charlie Liotta has proposed the following set of reactions to explain the prebiotic formation of complex sugars. Fill in the missing compounds.



7. **12 pts.** Provide the structure of the neutral polymer formed in each of the reactions below. Include the structure of the repeating unit in brackets with a subscript n ; do not include end groups and don't worry about stereochemistry.

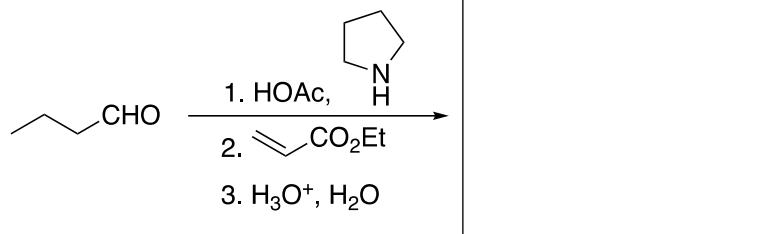


8. **6 pts.** Draw a block copolymer of monomers **A** and **B** in the box at right. Include at least ten repeating units total. An example of a homopolymer of **B** is below.

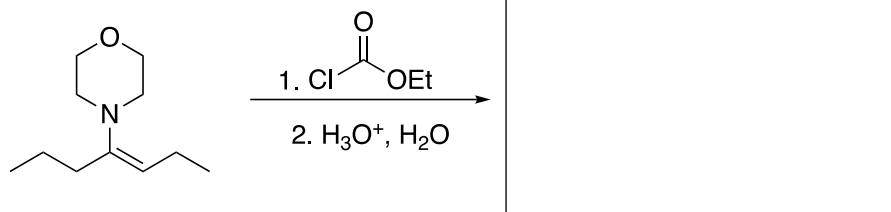


9. **20 pts.** Predict the neutral organic product of each reaction below.

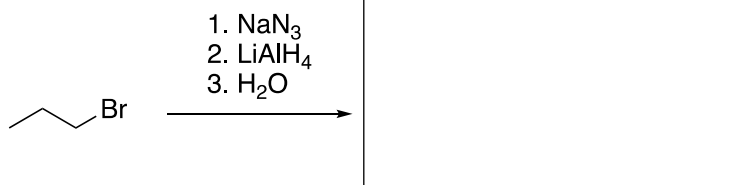
A.



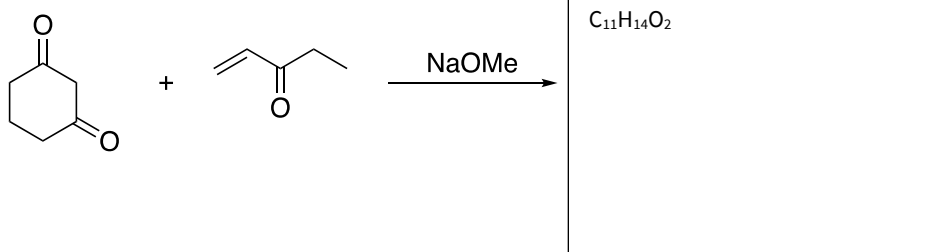
B.



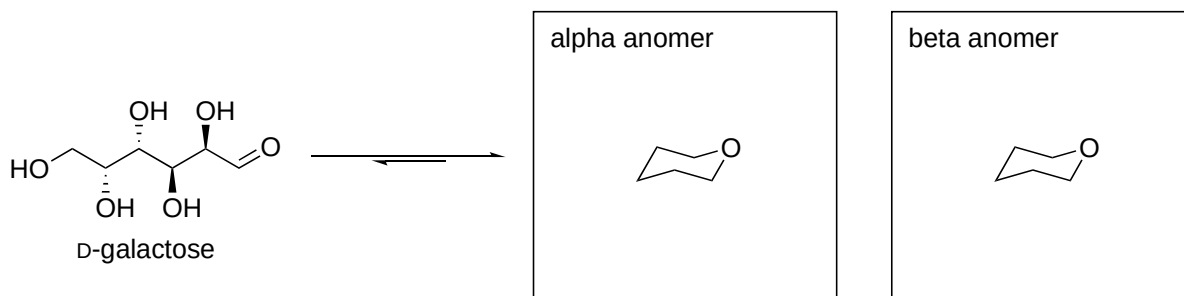
C.



D.



10. **12 pts.** In the boxes below, draw both anomers of the pyranose forms of D-galactose as their most stable chair conformers. Add -OH and -CH₂OH groups in the appropriate positions on the chair structures provided.



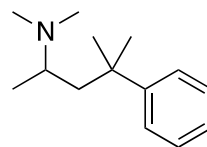
11. **12 pts.** In the boxes below, draw a pair of enantiomeric tetrose sugars (left-hand box) and diastereomeric tetrose sugars (right-hand box) as Fischer projections.

Enantiomeric tetrose sugars (two Fischer projections)

Diastereomeric tetrose sugars (two Fischer projections)

12. **10 pts.** Choose **one** of the two problems below, placing an X in the small gray box of your choice. Design a synthesis of the target (right) from the indicated starting material (left). You are welcome to use any existing reagents and sources of carbon to accomplish the synthesis, but it must start from the given starting materials.

A.



B.

